In [19]:
```
# --- Imports ---
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import unicodedata
import re
import datetime
import networkx as nx
from itertools import combinations
import pycountry
import plotly.graph_objects as go
```

Loading and Cleaning

In [20]:
```
# --- Data Loading & Cleaning for Flattened OpenAlex ---
def load_and_clean_your_csv(file_path):
    """
    Cleans the flattened OpenAlex CSV structure found in your research01 file.
    """
    df = pd.read_csv(file_path)

    # 1. Standardize Numeric Columns
    df['publication_year'] = pd.to_numeric(df['publication_year'], errors='coe
    df['cited_by_count'] = pd.to_numeric(df['cited_by_count'], errors='coerce'

    # 2. Extract Primary Author (takes the first name before the first '|')
    def get_primary_author(name_str):
        if pd.isna(name_str): return "Unknown"
        return str(name_str).split('|')[0]

    # 3. Handle Topics/Keywords (using the 'primary_topic.display_name' column
    # Note: If you want to use the full topic string, we keep it as is.
    df['primary_author'] = df['authorships.author.display_name'].apply(get_prin

    return df

# Load your specific file
df_clean = load_and_clean_your_csv('research01_works-csv-epopDs4cfD3FyuN8piis3A
```

In [21]:
```
df_clean.head()
```

Out[21]:

	id	authorships.author.display_name	authorsh
0	https://openalex.org/W2109101401	David Pisinger|Stefan Røpke	https://openalex.org/A5081910021|h
1	https://openalex.org/W2116675171	Olli Bräysy|Michel Gendreau	https://openalex.org/A5014631682|h
2	https://openalex.org/W2158000627	J-F Cordeau|G Laporte|Aurelién Mercier	https://openalex.org/A5059523142|h
3	https://openalex.org/W2084977713	Éric D. Taillard|Philippe Badeau|Michel Gendre...	https://openalex.org/A5071782451|h
4	https://openalex.org/W1975828823	Jean-Fran﹐ois Cordeau|Michel Gendreau|Gilbert ...	https://openalex.org/A5008962425|h

5 rows × 29 columns

In [22]:

```
# --- Rigorous cleaning (accents and extrange characters) --- #

def rigorous_clean(text):
    if not isinstance(text, str):
        return text

    # 1. Remove the '' and non-printable artifacts
    # This encodes to ASCII and ignores anything it can't translate
    text = text.encode("ascii", "ignore").decode("ascii")

    # 2. Normalize to remove accents correctly
    nksel = unicodedata.normalize('NFKD', text)
    clean_text = "".join([c for c in nksel if not unicodedata.combining(c)])

    # 3. Use Regex to keep only letters, numbers, pipes (for your separators),
    # This removes the "weird elements" and extra spaces between letters
    clean_text = re.sub(r'[^a-zA-Z0-9\s|.\-]', '', clean_text)

    # 4. Collapse multiple spaces into one
    clean_text = re.sub(r'\s+', ' ', clean_text).strip()

    return clean_text

# Apply to your columns
cols_to_fix = ['authorships.author.display_name', 'display_name', 'primary_topi

for col in cols_to_fix:
    if col in df_clean.columns:
        df_clean[col] = df_clean[col].apply(rigorous_clean)

print("Rigorous cleaning finished. No more squares or extra spaces.")
```

```
Rigorous cleaning finished. No more squares or extra spaces.
```

In [23]:

```
df_clean.head()
```

Out[23]:

	id	authorships.author.display_name	authorsh
0	https://openalex.org/W2109101401	David Pisinger|Stefan Rpke	https://openalex.org/A5081910021|h
1	https://openalex.org/W2116675171	Olli Brysy|Michel Gendreau	https://openalex.org/A5014631682|h
2	https://openalex.org/W2158000627	J-F Cordeau|G Laporte|Aurelin Mercier	https://openalex.org/A5059523142|h
3	https://openalex.org/W2084977713	ric D. Taillard|Philippe Badeau|Michel Gendrea...	https://openalex.org/A5071782451|h
4	https://openalex.org/W1975828823	Jean-Franois Cordeau|Michel Gendreau|Gilbert L...	https://openalex.org/A5008962425|h

5 rows × 29 columns

Function Creation

In [24]:

```
# --- Production Over Time ---
def plot_production_trends(df):
    # Filter years to match your study range (e.g., 2015-2025)
    yearly_counts = df[df['publication_year'] >= 2000]['publication_year'].valu

    plt.figure(figsize=(12, 6))
    sns.barplot(x=yearly_counts.index.astype(int), y=yearly_counts.values, colo
    plt.title('Documents per Year (VRP Review)', fontsize=16)
    plt.xlabel('Year', fontsize=14)
    plt.ylabel('Number of Documents', fontsize=14)
    plt.xticks(fontsize=12) # Increases size of the numbers on the X-axis
    plt.yticks(fontsize=12) # Increases size of the Author Names on the Y-axis
    plt.xticks(rotation=45)
    plt.grid(axis='y', linestyle='--', alpha=0.7)
    plt.show()
```

In [25]:

```
## -- Get country name ---##
def get_country_name(code):
    try:
        # Translates 'MX' -> 'Mexico', 'CN' -> 'China', etc.
        return pycountry.countries.get(alpha_2=code).name
    except:
        return code # Returns original code if translation fails
```

In [26]:

```
# --- Filtered Topic Analysis ---
def plot_real_trends(df):
    """
    Excludes the broad search terms to see the underlying application
    areas for Hybrid Augmented Intelligence.
    """
    # 1. Define the terms to remove (VRP and its obvious synonyms)
    exclude_list = [
        'vehicle routing problem',
        'vrp',
        'vehicle routing optimization methods',
        'routing optimization'
```

```
    ]

    # 2. Get the value counts of the primary topic column
    topic_counts = df['primary_topic.display_name'].value_counts()

    # 3. Filter the series using a negative regex match
    # This keeps only topics that DO NOT contain our exclusion terms
    pattern = '|'.join(exclude_list)
    filtered_topics = topic_counts[~topic_counts.index.str.contains(pattern, ca

    # 4. Grab the top 10 for the visualization
    top_10_real = filtered_topics.head(10)

    # 5. Visualization
    plt.figure(figsize=(15, 6))
    sns.barplot(x=top_10_real.values, y=top_10_real.index, palette='magma')
    plt.title('Top 10 Research Domains (VRP Terms Removed)', fontsize=14)
    plt.xlabel('Number of Articles')
    plt.ylabel('')
    plt.xticks(fontsize=12) # Increases size of the numbers on the X-axis
    plt.yticks(fontsize=12) # Increases size of the Author Names on the Y-axis
    plt.tight_layout()
    plt.show()
```

In [27]:

```
# --- Top Cited Articles (Identification of Influential Works) ---
def show_top_cited(df):
    """Shows the most influential papers for your review."""
    top_papers = df[['display_name', 'primary_author', 'publication_year', 'cit
    return top_papers
```

In [28]:

```
##--- Plot top stats ----

def plot_top_stats(df, category='author', metric='publications', top_n=10,
                   citation_col='cited_by_count', separator='|'):
    """
    Unified function adjusted for your specific research01 dataset column names
    """

    # 1. Map generic categories to YOUR specific DataFrame column names
    column_map = {
        'author': 'authorships.author.display_name',
        'institution': 'authorships.institutions.display_name',
        'country': 'authorships.countries'
    }

    target_col = column_map.get(category.lower())

    if not target_col or target_col not in df.columns:
        print(f"Error: Category '{category}' invalid or column '{target_col}' n
        return

    # 2. Pre-process: Copy only needed columns
    cols_needed = [target_col]
    if metric == 'citations':
        cols_needed.append(citation_col)
    df_temp = df[cols_needed].copy()

    # 3. Clean and Explode (Split values by the pipe '|')
    df_temp[target_col] = df_temp[target_col].fillna('').astype(str).str.split(
```

```
    exploded = df_temp.explode(target_col)

    # 4. Rigorous Cleaning
    if category == 'author':
        exploded[target_col] = exploded[target_col].apply(rigorous_clean)
    elif category == 'country':
        # Translate codes to names
        exploded[target_col] = exploded[target_col].str.strip().apply(get_count
    else:
        exploded[target_col] = exploded[target_col].str.strip()

    # Remove garbage strings (length <= 1)
    exploded = exploded[exploded[target_col].str.len() > 1]

    # 5. Calculation Logic
    if metric == 'publications':
        top_data = exploded.groupby(target_col).size().sort_values(ascending=Fa
        title_text = f'Top {top_n} {category.capitalize()}s (by Publication Cou
        x_label = 'Number of Documents'
        palette = 'viridis'

    elif metric == 'citations':
        top_data = exploded.groupby(target_col)[citation_col].sum().sort_values
        title_text = f'Top {top_n} {category.capitalize()}s (by Total Citations
        x_label = 'Total Citations'
        palette = 'rocket'

    # 6. Visualization
    plt.figure(figsize=(12, 8))
    sns.barplot(x=top_data.values, y=top_data.index, palette=palette)

    plt.title(title_text, fontsize=16)
    plt.xlabel(x_label, fontsize=14)
    #plt.ylabel(category.capitalize(), fontsize=14)
    plt.ylabel('')
    plt.xticks(fontsize=12)
    plt.yticks(fontsize=12)
    plt.tight_layout()
    plt.show()

    return top_data
```

In [29]:

```
# --- Filtered database by the last n years ----#
def filter_by_last_n_years(df, n_years):
    """
    Filters the dataframe to include only articles from the last n_years.
    Useful for defining the '2015-2025' timeframe for your review.
    """
    # 1. Ensure publication_year is numeric
    df['publication_year'] = pd.to_numeric(df['publication_year'], errors='coer

    # 2. Determine the reference year (Current year is 2026)
    current_year = 2026
    start_year = current_year - n_years

    # 3. Create the filtered dataframe
    filtered_df = df[df['publication_year'] >= start_year].copy()

    print(f"Original articles: {len(df)}")
```

```
    print(f"Articles from {start_year} to {current_year}: {len(filtered_df)}")

    return filtered_df
```

In [30]:

```
## --- Author clusters ----#
def analyze_author_clusters(df, authors_column, min_collaborations=4, separato
                            grid_rows=2, grid_cols=2, node_size=500, font_size=
                            edge_width_scale=0.5):
    """
    Plots the top author clusters in a grid layout.

    Parameters:
    - df: DataFrame containing the data.
    - authors_column: Name of the column with authors.
    - min_collaborations: Minimum shared papers to create a link.
    - separator: Character separating authors (e.g., '|' or ';').
    - grid_rows, grid_cols: Dimensions of the subplot grid (e.g., 2x2 = top 4 c
    - node_size: Visual size of the author bubbles.
    - font_size: Size of the author names.
    - edge_width_scale: Multiplier for edge thickness.
    """

    # --- 1. BUILD THE GRAPH ---
    G = nx.Graph()

    # Iterate through publications
    for authors_str in df[authors_column].dropna():
        # Split and clean author names
        authors = [a.strip() for a in str(authors_str).split(separator) if a.st

        # Create edges for every pair of co-authors
        if len(authors) > 1:
            for u, v in combinations(authors, 2):
                if G.has_edge(u, v):
                    G[u][v]['weight'] += 1
                else:
                    G.add_edge(u, v, weight=1)

    # --- 2. FILTER & SORT CLUSTERS ---
    # Keep only edges that meet the minimum collaboration threshold
    strong_edges = [(u, v) for u, v, d in G.edges(data=True) if d['weight'] >=
    G_filtered = G.edge_subgraph(strong_edges).copy()

    # Get all connected components (clusters) and sort them by size (largest fi
    # This ensures we prioritize the "most collaborative" groups
    clusters = sorted(nx.connected_components(G_filtered), key=len, reverse=Tru

    if not clusters:
        print(f"No collaborations found with >= {min_collaborations} shared pap
        return

    # --- 3. CREATE THE GRID PLOT ---
    num_plots = grid_rows * grid_cols
    fig, axes = plt.subplots(grid_rows, grid_cols, figsize=(grid_cols * 6, gric

    # Flatten axes array for easy iteration (handles 1x1, 1xN, NxN cases)
    if grid_rows * grid_cols > 1:
        axes = axes.flatten()
    else:
```

```
        axes = [axes]

    print(f"Found {len(clusters)} clusters. Showing the top {min(len(clusters),

    for i in range(num_plots):
        ax = axes[i]

        # If we run out of clusters, hide the empty subplot
        if i >= len(clusters):
            ax.axis('off')
            continue

        # Get the subgraph for this specific cluster
        cluster_nodes = clusters[i]
        sub_G = G_filtered.subgraph(cluster_nodes)

        # Layout: distinct shape for each cluster
        pos = nx.spring_layout(sub_G, k=0.5, seed=42)

        # Dynamic edge thickness based on collaboration count
        weights = [sub_G[u][v]['weight'] for u, v in sub_G.edges()]
        widths = [w * edge_width_scale for w in weights]

        # Draw the plot
        nx.draw_networkx_nodes(sub_G, pos, ax=ax, node_size=node_size, node_col
        nx.draw_networkx_edges(sub_G, pos, ax=ax, width=widths, alpha=0.5, edge
        nx.draw_networkx_labels(sub_G, pos, ax=ax, font_size=font_size, font_fa

        ax.set_title(f"Cluster #{i+1} ({len(cluster_nodes)} Authors)", fontsize
        ax.axis('off')

    plt.tight_layout()
    plt.show()

# --- EXAMPLE USAGE ---
# analyze_author_clusters(
#     df,
#     authors_column='Authors',
#     min_collaborations=5,
#     separator='|',
#     grid_rows=2,
#     grid_cols=2,   # This will show the top 4 biggest groups
#     node_size=600,
#     font_size=8
# )
```

In [31]:

```
## --- Colaboration clusters --- ##
def analyze_category_clusters(df, category='author', min_shared=4,
                              grid_rows=2, grid_cols=2, node_size=20,
                              font_size=12, edge_width_scale=0.5):
    """
    Generalized function to plot collaboration clusters for Authors, Institutio
    """

    # 1. Map categories to your specific CSV columns
    column_map = {
        'author': 'authorships.author.display_name',
        'institution': 'authorships.institutions.display_name',
        'country': 'authorships.countries'
```

```
    }

    target_col = column_map.get(category.lower())
    if not target_col:
        print(f"Error: Category '{category}' not supported.")
        return

    # 2. Country name translation helper
    def translate_country(code):
        try: return pycountry.countries.get(alpha_2=code.strip()).name
        except: return code.strip()

    # 3. Build the Graph
    G = nx.Graph()

    for items_str in df[target_col].dropna():
        # Split and clean
        items = [i.strip() for i in str(items_str).split('|') if i.strip()]

        # Category-specific cleaning
        if category == 'author':
            items = [rigorous_clean(i) for i in items]
        elif category == 'country':
            items = [translate_country(i) for i in items]

        # Get UNIQUE items per paper to find true COLLABORATION (e.g. Universi
        unique_items = list(set(items))

        if len(unique_items) > 1:
            for u, v in combinations(unique_items, 2):
                if G.has_edge(u, v):
                    G[u][v]['weight'] += 1
                else:
                    G.add_edge(u, v, weight=1)

    # 4. Filter strong connections & find clusters
    strong_edges = [(u, v) for u, v, d in G.edges(data=True) if d['weight'] >=
    G_filtered = G.edge_subgraph(strong_edges).copy()
    clusters = sorted(nx.connected_components(G_filtered), key=len, reverse=Tru

    if not clusters:
        print(f"No {category} clusters found with >= {min_shared} shared papers
        return

    # 5. Plot Grid
    num_plots = grid_rows * grid_cols
    fig, axes = plt.subplots(grid_rows, grid_cols, figsize=(grid_cols * 6, gric
    axes = axes.flatten() if num_plots > 1 else [axes]

    print(f"Showing top {min(len(clusters), num_plots)} {category} clusters.")

    for i in range(num_plots):
        ax = axes[i]
        if i >= len(clusters):
            ax.axis('off')
            continue

        sub_G = G_filtered.subgraph(clusters[i])
        pos = nx.spring_layout(sub_G, k=0.6, seed=42)
```

```
        # Scale edge width by weight
        widths = [sub_G[u][v]['weight'] * edge_width_scale for u, v in sub_G.ec

        nx.draw_networkx_nodes(sub_G, pos, ax=ax, node_size=node_size, node_col
        nx.draw_networkx_edges(sub_G, pos, ax=ax, width=widths, alpha=0.4, edge

        # Dynamic label positioning
        labels = {node: node.replace(' ', '\n') for node in sub_G.nodes()}
        nx.draw_networkx_labels(sub_G, pos, labels=labels, ax=ax, font_size=for

        ax.set_title(f"{category.capitalize()} Cluster #{i+1} ({len(clusters[i]
        ax.axis('off')

    plt.tight_layout()
    plt.show()
```

In [32]:

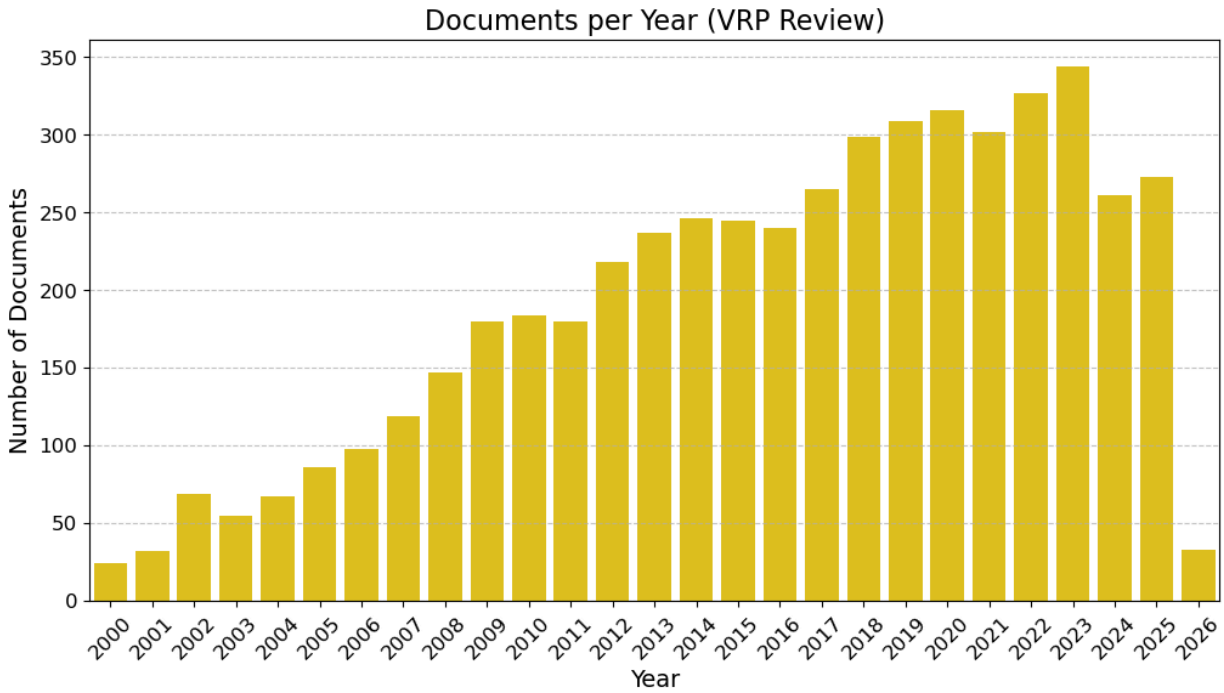
```
## --- Sankey diagrams ---##

def plot_sankey_categories(df, categories=['country', 'topic', 'author'], n_ite
    # 1. Column Mapping (Specific to your research01.csv)
    column_map = {
        'author': 'authorships.author.display_name',
        'institution': 'authorships.institutions.display_name',
        'country': 'authorships.countries',
        'topic': 'primary_topic.display_name'
    }

    # 2. Filtering Logic (VRP Exclusions)
    exclude_vrp = ['vehicle routing problem', 'vrp', 'optimization methods', 'i
    pattern = '|'.join(exclude_vrp)

    def get_country_name(code):
        try: return pycountry.countries.get(alpha_2=code).name
        except: return code

    # 3. Processing
    cols = [column_map[c] for c in categories]
    temp_df = df[cols].dropna().copy()

    for cat in categories:
        col = column_map[cat]
        temp_df[col] = temp_df[col].astype(str).str.split('|')
        temp_df = temp_df.explode(col)
        temp_df[col] = temp_df[col].str.strip()

        if cat == 'author':
            temp_df[col] = temp_df[col].apply(rigorous_clean)
        elif cat == 'country':
            temp_df[col] = temp_df[col].apply(get_country_name)
        elif cat == 'topic':
            temp_df = temp_df[~temp_df[col].str.contains(pattern, case=False, r
            # TRUNCATE LONG TOPICS for visual clarity
            temp_df[col] = temp_df[col].apply(lambda x: (x[:25] + '...') if len

    # Filter for top N to avoid "horrible" clutter
    for cat in categories:
        col = column_map[cat]
        top_items = temp_df[col].value_counts().head(n_items).index
        temp_df = temp_df[temp_df[col].isin(top_items)]
```

```
    # 4. Create Links
    links = []
    for i in range(len(categories) - 1):
        source_col = column_map[categories[i]]
        target_col = column_map[categories[i+1]]
        res = temp_df.groupby([source_col, target_col]).size().reset_index(name
        res.columns = ['source', 'target', 'value']
        links.append(res)

    sankey_df = pd.concat(links)
    all_nodes = list(pd.concat([sankey_df['source'], sankey_df['target']]).unic
    node_map = {node: i for i, node in enumerate(all_nodes)}

    # 5. Build the Chart with specific "Clean" styling
    fig = go.Figure(data=[go.Sankey(
        node=dict(
            pad=30,           # More space between boxes
            thickness=15,     # Thinner, more elegant boxes
            line=dict(color="gray", width=0.5),
            label=all_nodes,
            color="royalblue" # Uniform node color for academic look
        ),
        link=dict(
            source=sankey_df['source'].map(node_map),
            target=sankey_df['target'].map(node_map),
            value=sankey_df['value'],
            color="rgba(173, 216, 230, 0.4)" # Light blue transparent links
        )
    )])

    fig.update_layout(
        title_text=f"Mapping the Scientific Landscape: {' > '.join(categories).
        font_size=14,
        height=700,
        margin=dict(l=50, r=50, t=80, b=50)
    )
    fig.show()

# Run it with a smaller n_items for a cleaner look
# plot_clean_sankey(df_recent, categories=['country', 'topic', 'author'], n_ite
```

General Results

Production Over Time

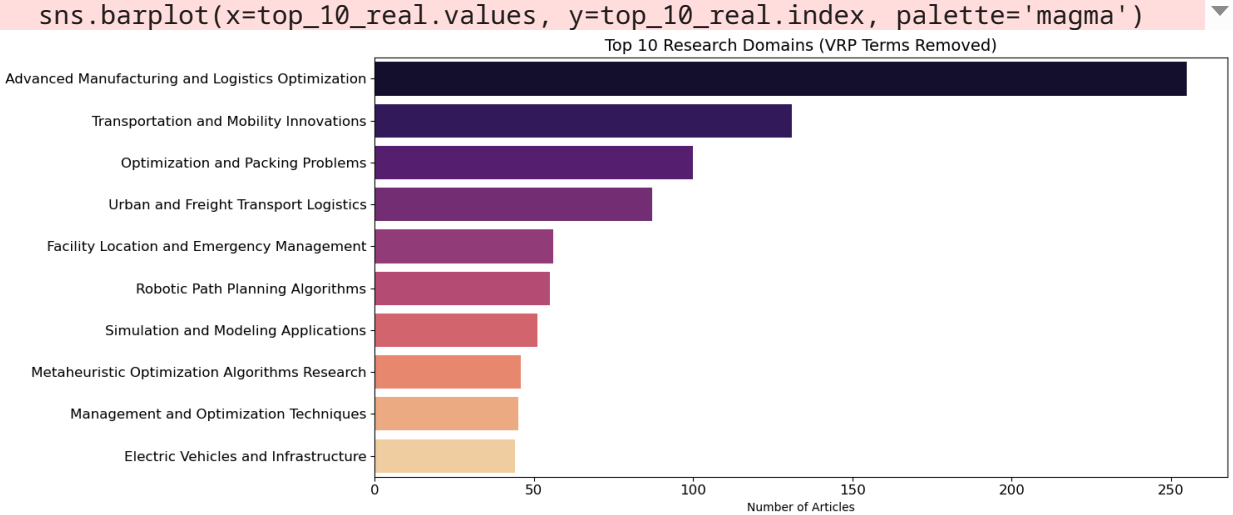
In [33]:
```
plot_production_trends(df_clean)
```



Topic Analysis

In [34]:
```
plot_real_trends(df_clean)
```

```
/tmp/ipykernel_1554987/2377348178.py:28: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_10_real.values, y=top_10_real.index, palette='magma')
```



Top Cited Articles

In [35]:
```
show_top_cited(df_clean)
```

Out[35]:

	display_name	primary_author	publication_year	cited_by_count
153	The Vehicle Routing Problem	Paolo Toth	2002.0	3958
430	An Adaptive Large Neighborhood Search Heuristi...	Stefan Røpke	2006.0	2241
387	The vehicle routing problem An overview of exa...	Gilbert Laporte	1992.0	1705
696	The flying sidekick traveling salesman problem...	Chase Murray	2015.0	1464
0	A general heuristic for vehicle routing problems	David Pisinger	2005.0	1360
513	Using Constraint Programming and Local Search ...	Paul Shaw	1998.0	1282
1042	Handbook of Scheduling Algorithms Models and P...	Joseph Leung	2004.0	1225
603	The Electric Vehicle-Routing Problem with Time...	Michael Schneider	2014.0	1178
611	Vehicle Routing Problems for Drone Delivery	Kevin Dorling	2016.0	1173
359	The multiple traveling salesman problem an ove...	Tolga Bektaş	2005.0	1119

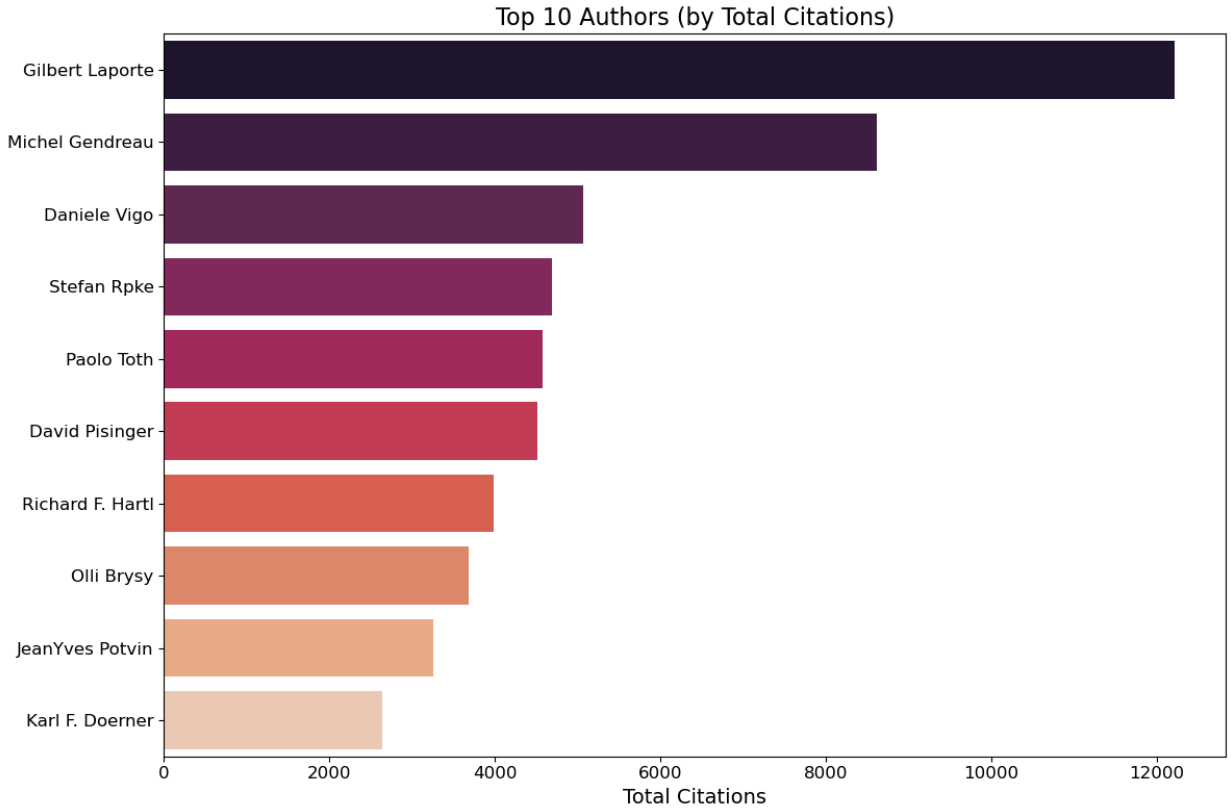
Top Authors

Per citations

In [37]:

```
plot_top_stats(df_clean,category = 'author', metric='citations')
```

```
/tmp/ipykernel_1554987/2788011657.py:59: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_data.values, y=top_data.index, palette=palette)
```



Out[37]:

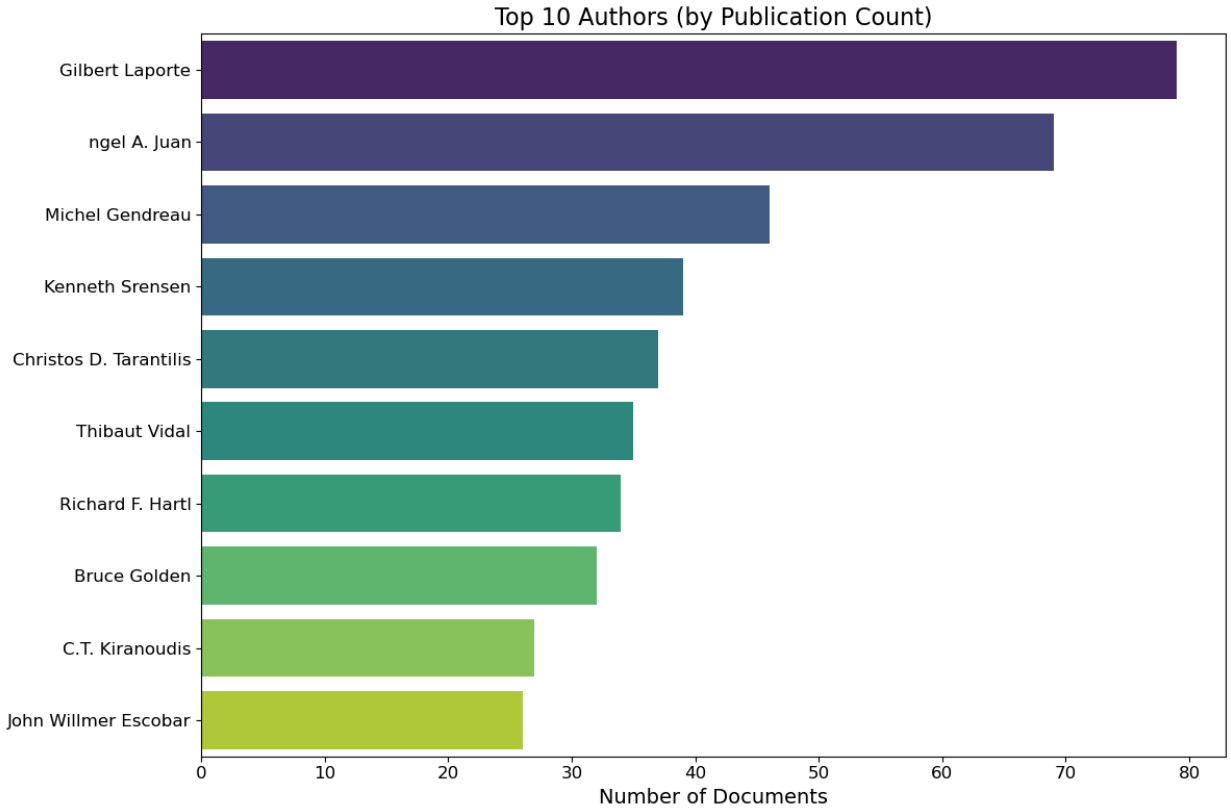
```
authorships.author.display_name
Gilbert Laporte     12219
Michel Gendreau      8616
Daniele Vigo         5064
Stefan Rpke          4693
Paolo Toth           4570
David Pisinger       4514
Richard F. Hartl     3982
Olli Brysy           3688
JeanYves Potvin      3251
Karl F. Doerner      2644
Name: cited_by_count, dtype: int64
```

Per publications

In [38]:

```
plot_top_stats(df_clean,category = 'author', metric='publications')
```

```
/tmp/ipykernel_1554987/2788011657.py:59: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_data.values, y=top_data.index, palette=palette)
```



Out[38]:

```
authorships.author.display_name
Gilbert Laporte           79
ngel A. Juan              69
Michel Gendreau           46
Kenneth Srensen           39
Christos D. Tarantilis    37
Thibaut Vidal             35
Richard F. Hartl          34
Bruce Golden              32
C.T. Kiranoudis           27
John Willmer Escobar      26
dtype: int64
```

Last 10 Years Results

In [ ]:

```
df_recent = filter_by_last_n_years(df_clean, 11)

# Verify the range
print(f"Minimum year in filtered set: {df_recent['publication_year'].min()}")
```

```
Original articles: 5422
Articles from 2015 to 2026: 3214
Minimum year in filtered set: 2015.0
```

In [ ]:

```
df_recent.head()
```

Out[ ]:

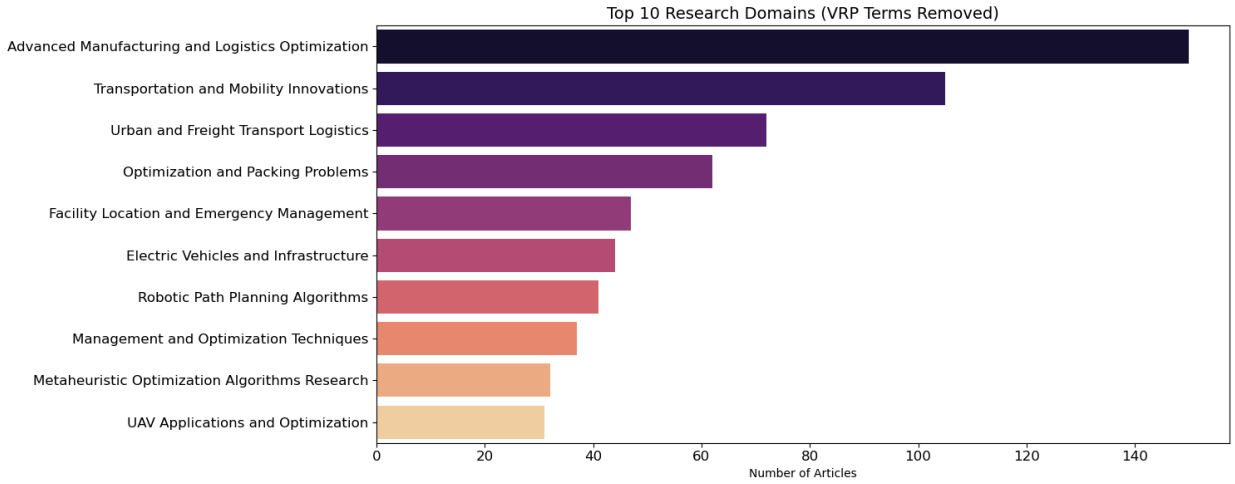
	id	authorships.author.display_name	authors
7	https://openalex.org/W2927378423	David Sacramento|David Pisinger|Stefan Rpke	https://openalex.org/A5041840487|
12	https://openalex.org/W3136908367	Wei Qin|Zilong Zhuang|Zizhao Huang|Haozhe Huang	https://openalex.org/A5054830882|
13	https://openalex.org/W2996796639	Raafat Elshaer|Hadeer Awad	https://openalex.org/A5004291940|
16	https://openalex.org/W3111551443	Bra Olgun|ar Ko|Fulya Altparmak	https://openalex.org/A5004938821|
34	https://openalex.org/W2226731518	ar Ko|smail Karaolan	https://openalex.org/A5058877968|

5 rows × 29 columns

Topic Analysis

In [ ]:
```
plot_real_trends(df_recent)
```

```
/tmp/ipykernel_1480527/2377348178.py:28: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_10_real.values, y=top_10_real.index, palette='magma')
```



Top Cited Articles

In [ ]:
```
show_top_cited(df_recent)
```

Out[ ]:

	display_name	primary_author	publication_year	cited_by_count
696	The flying sidekick traveling salesman problem...	Chase Murray	2015.0	1464
611	Vehicle Routing Problems for Drone Delivery	Kevin Dorling	2016.0	1173
7	An adaptive large neighborhood search metaheur...	David Sacramento	2019.0	468
944	The Vehicle Routing Problem with Occasional Dr...	Claudia Archetti	2016.0	361
13	A taxonomic review of metaheuristic algorithms...	Raafat Elshaer	2019.0	283
691	A Survey on the Electric Vehicle Routing Probl...	Tomislav Erdelić	2019.0	251
794	Thirty years of heterogeneous vehicle routing	Çağrı Koç	2015.0	250
948	A hybrid VNSTabu search algorithm for solving ...	Daniel Schermer	2019.0	244
928	The electric vehicle routing problem with ener...	Samuel Pelletier	2019.0	239
1023	A parallel simulated annealing method for the ...	Chao Wang	2015.0	228

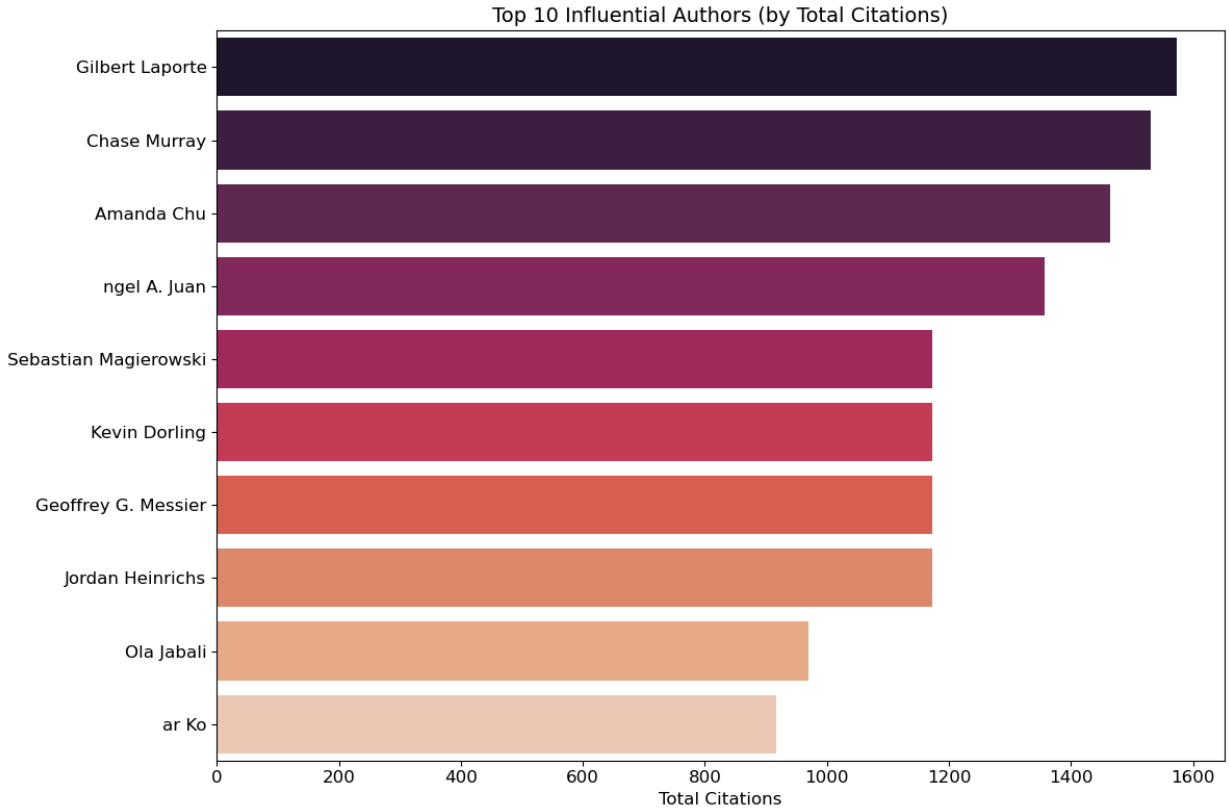
Top Authors

Per citations

In [ ]:

```
plot_top_stats(df_recent,category = 'author', metric='citations')
```

```
/tmp/ipykernel_1501292/3487523125.py:82: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_data.values, y=top_data.index, palette=palette)
```

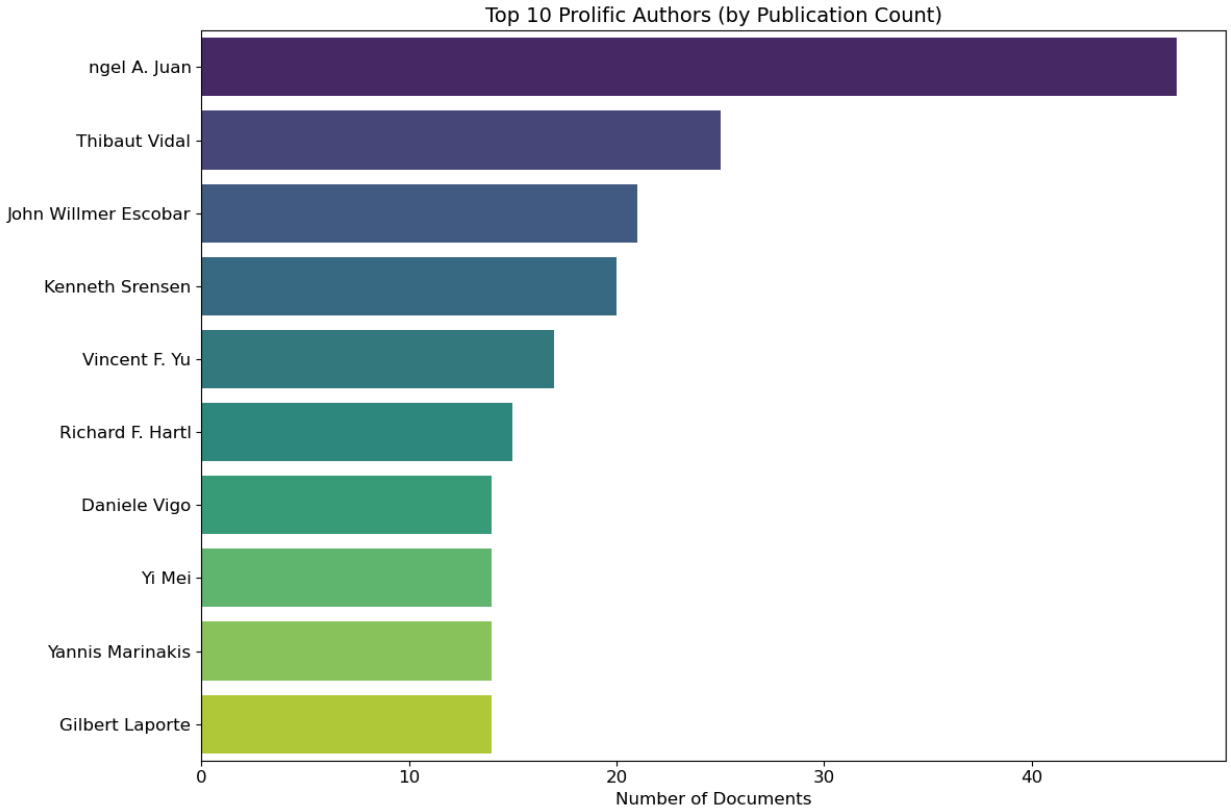


Out[ ]:
```
authorships.author.display_name
Gilbert Laporte          1573
Chase Murray             1531
Amanda Chu               1464
ngel A. Juan             1356
Sebastian Magierowski    1173
Kevin Dorling            1173
Geoffrey G. Messier      1173
Jordan Heinrichs         1173
Ola Jabali                970
ar Ko                     916
Name: cited_by_count, dtype: int64
```

Per publications

In [ ]:
```
plot_top_stats(df_recent,category = 'author',metric='publications')
```

```
/tmp/ipykernel_1501292/3487523125.py:82: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_data.values, y=top_data.index, palette=palette)
```



Out[ ]:

```
authorships.author.display_name
ngel A. Juan             47
Thibaut Vidal            25
John Willmer Escobar     21
Kenneth Srensen          20
Vincent F. Yu            17
Richard F. Hartl         15
Daniele Vigo             14
Yi Mei                   14
Yannis Marinakis         14
Gilbert Laporte          14
dtype: int64
```

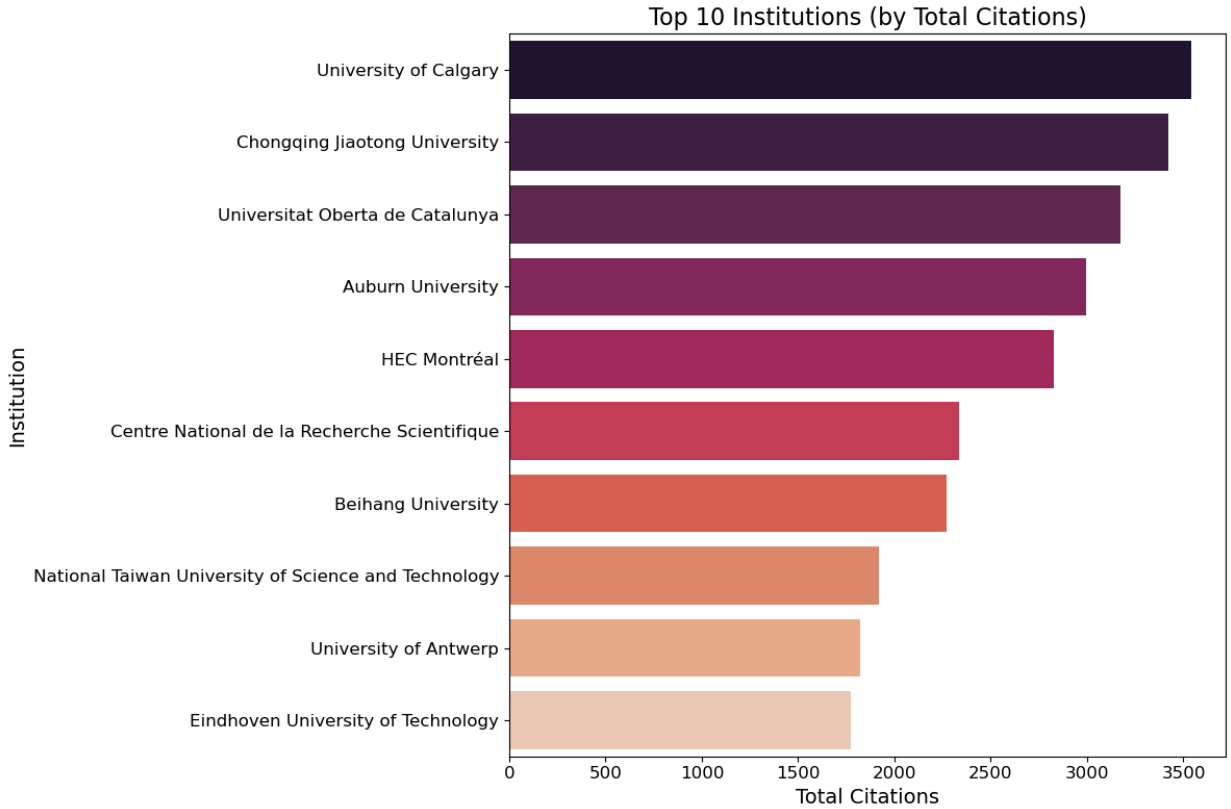
Top Institutions

Per citations

In [ ]:

```
plot_top_stats(df_recent,category = 'institution', metric='citations')
```

```
/tmp/ipykernel_1528290/3605809903.py:59: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_data.values, y=top_data.index, palette=palette)
```



Out[ ]:

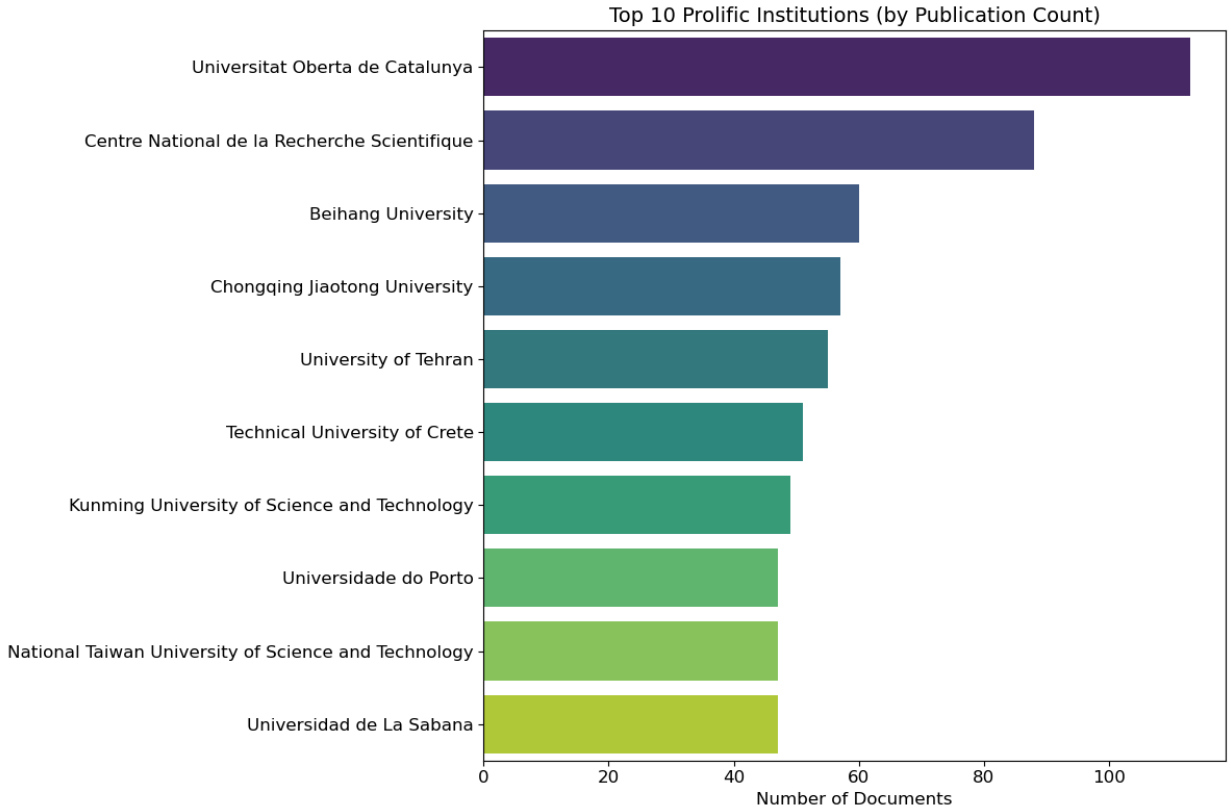
```
authorships.institutions.display_name
University of Calgary                                    3542
Chongqing Jiaotong University                            3423
Universitat Oberta de Catalunya                          3172
Auburn University                                        2995
HEC Montréal                                             2828
Centre National de la Recherche Scientifique             2333
Beihang University                                       2270
National Taiwan University of Science and Technology     1921
University of Antwerp                                    1824
Eindhoven University of Technology                       1775
Name: cited_by_count, dtype: int64
```

Per Publications

In [ ]:

```
plot_top_stats(df_recent,category = 'institution', metric='publications')
```

```
/tmp/ipykernel_1501292/3487523125.py:82: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_data.values, y=top_data.index, palette=palette)
```



Out[ ]:

```
authorships.institutions.display_name
Universitat Oberta de Catalunya                          113
Centre National de la Recherche Scientifique              88
Beihang University                                        60
Chongqing Jiaotong University                             57
University of Tehran                                      55
Technical University of Crete                             51
Kunming University of Science and Technology              49
Universidade do Porto                                     47
National Taiwan University of Science and Technology      47
Universidad de La Sabana                                  47
dtype: int64
```

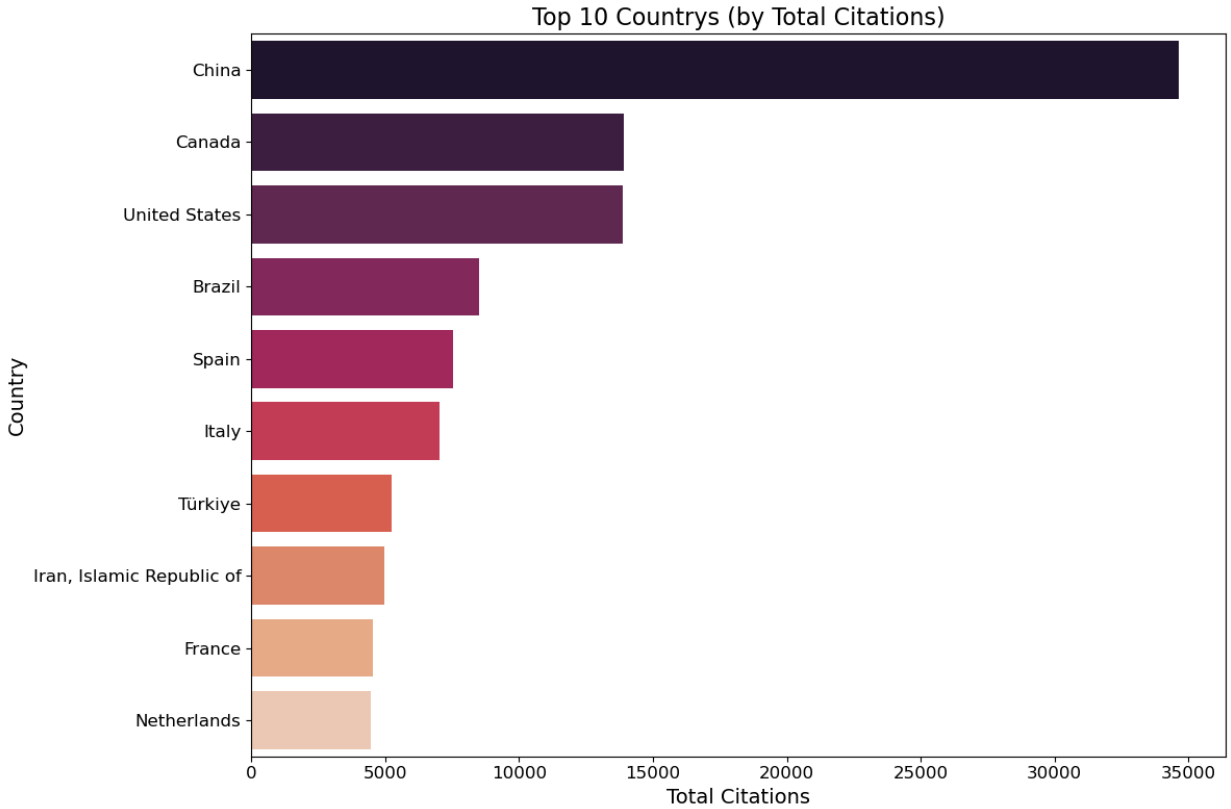
Top Countries

Per citations

In [ ]:

```
plot_top_stats(df_recent,category = 'country', metric='citations')
```

```
/tmp/ipykernel_1528290/3605809903.py:59: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_data.values, y=top_data.index, palette=palette)
```



Out[ ]:

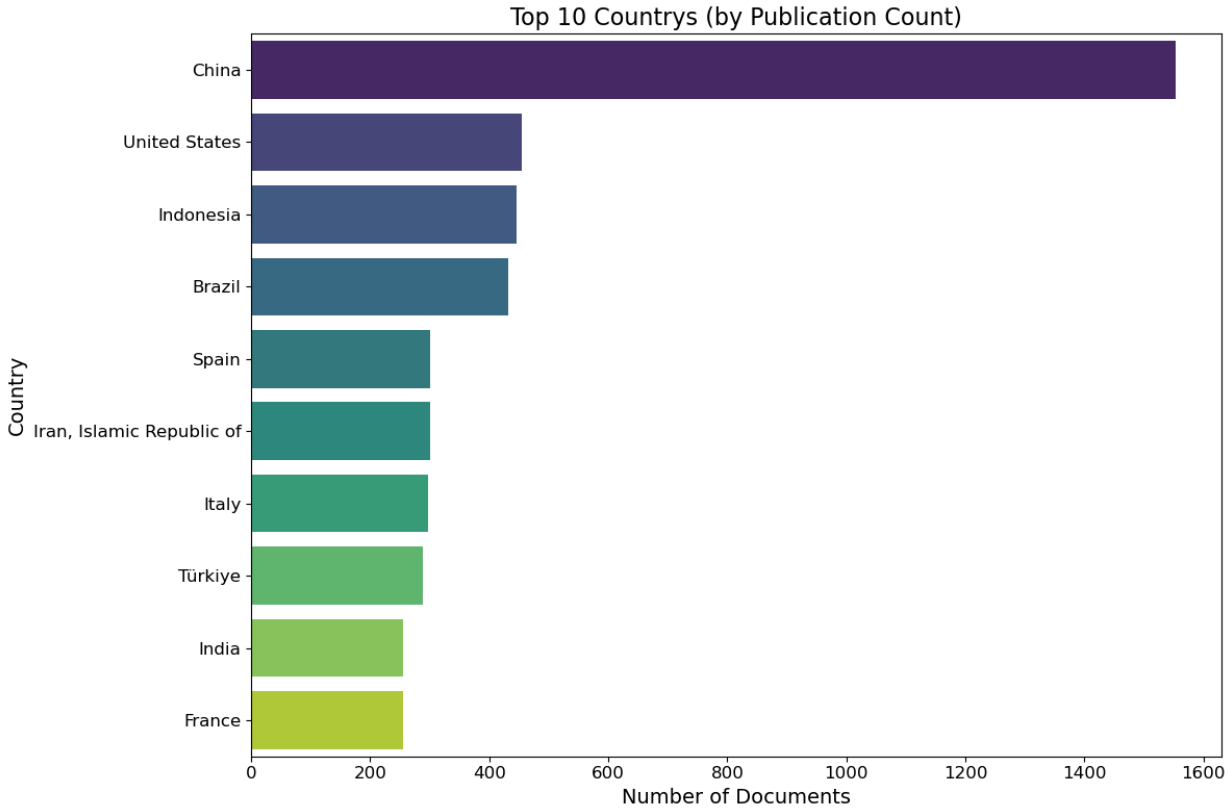
```
authorships.countries
China                        34655
Canada                       13895
United States                13856
Brazil                        8509
Spain                         7540
Italy                         7019
Türkiye                       5236
Iran, Islamic Republic of     4969
France                        4534
Netherlands                   4457
Name: cited_by_count, dtype: int64
```

Per publications

In [ ]:

```
plot_top_stats(df_recent,category = 'country', metric='publications')
```

```
/tmp/ipykernel_1528290/3605809903.py:59: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed
in v0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the
same effect.

  sns.barplot(x=top_data.values, y=top_data.index, palette=palette)
```



Out[ ]:

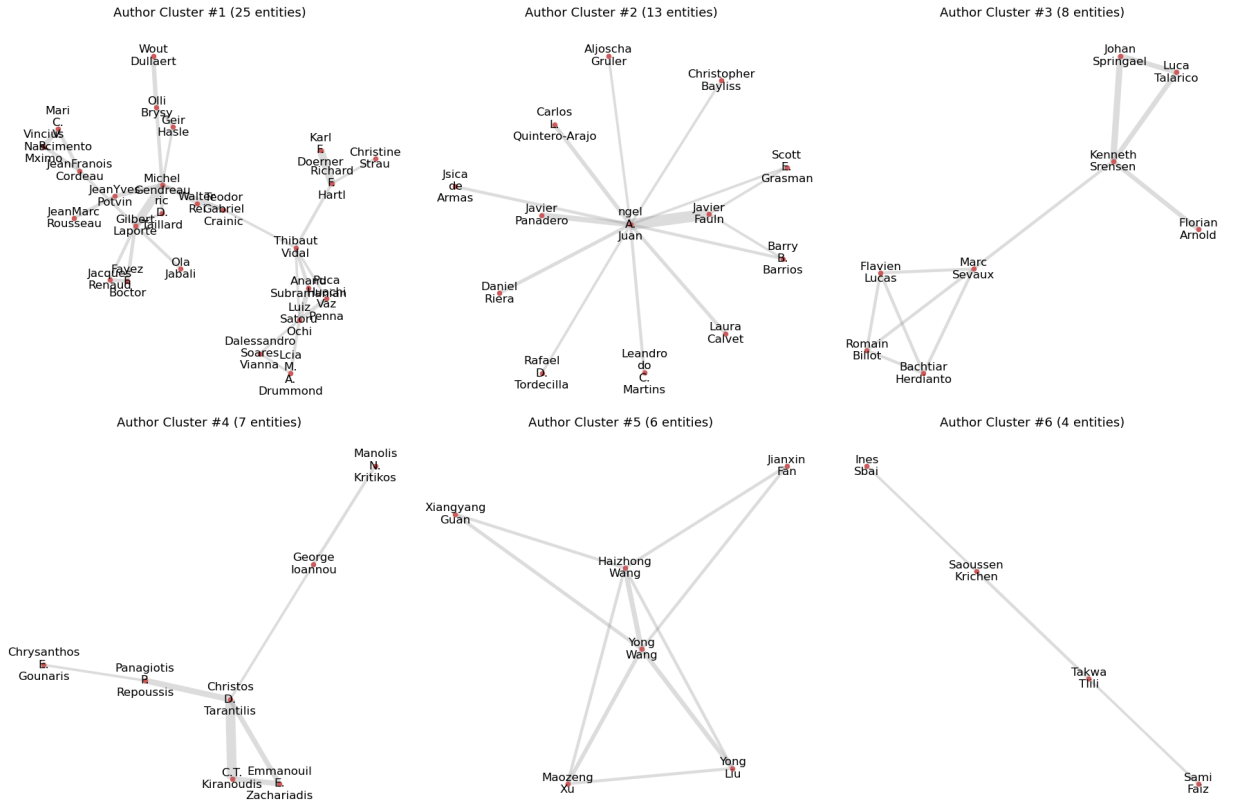
```
authorships.countries
China                        1553
United States                 455
Indonesia                     445
Brazil                        432
Spain                         301
Iran, Islamic Republic of     300
Italy                         297
Türkiye                       289
India                         256
France                        255
dtype: int64
```

Colaborations

In [ ]:

```
analyze_category_clusters(df_clean, category='author', min_shared=5, grid_rows=
```

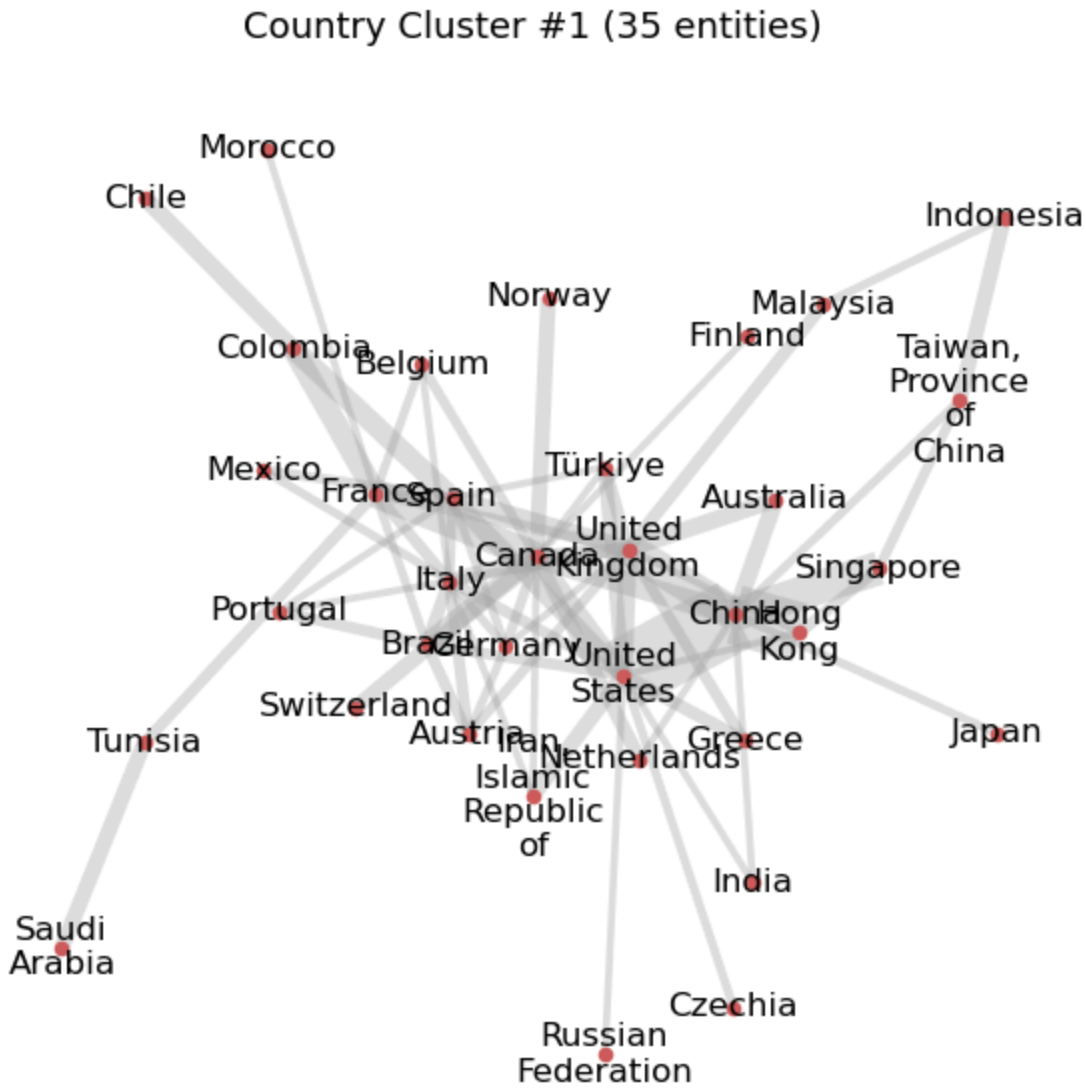
```
Showing top 6 author clusters.
```



In [ ]:

```
# --- Example Usages ---
analyze_category_clusters(df_clean, category='country', min_shared=5, grid_rows
```

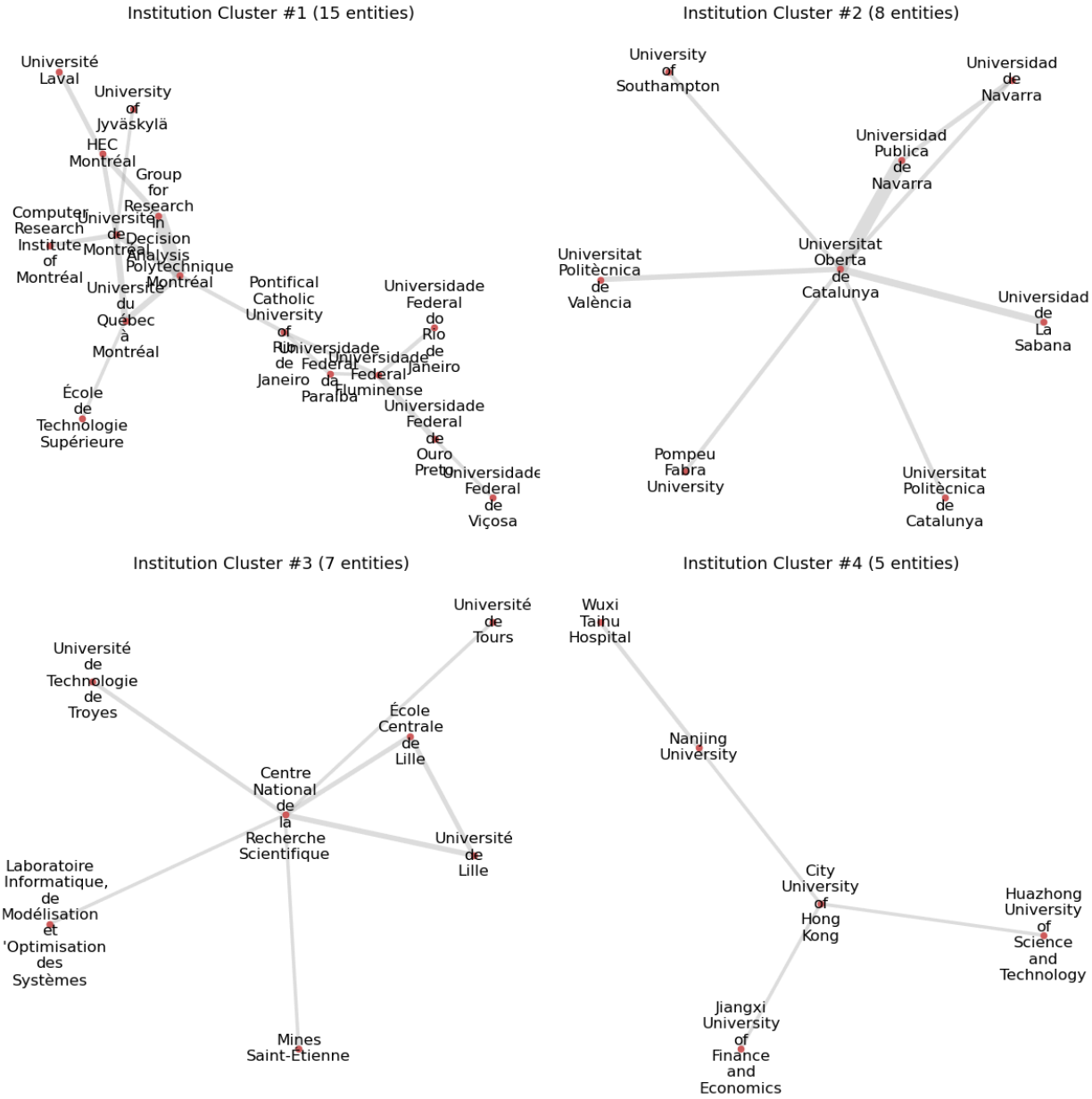
```
Showing top 1 country clusters.
```



In [ ]:
```
analyze_category_clusters(df_clean, category='institution', min_shared=5, grid_
```

```
Showing top 4 institution clusters.
```

```
/tmp/ipykernel_1528290/3356038994.py:78: DeprecationWarning:

`alltrue` is deprecated as of NumPy 1.25.0, and will be removed in NumPy 2.0.
Please use `all` instead.
```


Institution Cluster #1 (15 entities)
Université Laval
University of Jyväskylä
HEC Montréal
Group for Research in Decision Analysis
Computer Research Institute of Montréal
Université de Montréal
Polytechnique Montréal
Université du Québec à Montréal
École de Technologie Supérieure
Pontifical Catholic University of Rio de Janeiro
Universidade Federal do Rio de Janeiro
Universidade Federal da Paraíba
Universidade Federal Fluminense
Universidade Federal de Ouro Preto
Universidade Federal de Viçosa
Institution Cluster #2 (8 entities)
University of Southampton
Universidad de Navarra
Universidad Publica de Navarra
Universitat Politècnica de València
Universitat Oberta de Catalunya
Universidad de La Sabana
Pompeu Fabra University
Universitat Politècnica de Catalunya
Institution Cluster #3 (7 entities)
Université de Tours
Université de Technologie de Troyes
École Centrale de Lille
Centre National de la Recherche Scientifique
Université de Lille
Laboratoire Informatique, de Modélisation et 'Optimisation des Systèmes
Mines Saint-Etienne
Institution Cluster #4 (5 entities)
Wuxi Taihu Hospital
Nanjing University
City University of Hong Kong
Huazhong University of Science and Technology
Jiangxi University of Finance and Economics

In [ ]: